

A scalable and long-cycle-life 600 Wh kg⁻¹ solid-state lithium metal pouch cell

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Emerging applications, such as electric aircraft and ships, demand rechargeable batteries with high specific energy. However, current lithium (Li) ion batteries fall significantly short in this regard. Even with Li metal batteries, achieving a high cell-level specific energy (e.g. ≥ 600 Wh kg⁻¹) remains a significant challenge. In this study, a synergistic strategy integrating a precisely engineered carbonate-based gel-solid-state electrolyte with a surface-modified Li is proposed. Our in-situ gelation technique confines unstable free carbonate solvent molecules, thereby suppressing the electrolyte-induced parasitic reactions. Furthermore, a bicontinuous gradient polymer layer that regulates Li deposition and improves moisture/oxygen resistance is introduced, enabling damage-free Li processing and avoiding harsh production requirements. As a result, our 11 Ah-level Li metal battery achieves a high specific energy of 604.2 Wh kg⁻¹ (626.4 Wh kg⁻¹ excluding the packaging materials) and operates for over 100 cycles with an energy retention of 92.83% under lean electrolyte conditions (0.85 g Ah⁻¹). This work offers a practical strategy for the scalable production of long-cycle Li metal batteries with high specific energy for broad future applications.

Transportation electrification is crucial for decarbonizing the global economy, addressing the greenhouse effect, and achieving a sustainable energy transition. Diverse emerging transporting scenarios, such as battery-powered ships (BPSs), electric vertical take-off and landings (eVTOLs), and unmanned aerial vehicles (UAVs), are on the verge of widespread application. However, state-of-the-art Li-ion batteries deliver specific energies that peak around 360 Wh kg⁻¹, falling short of the minimum requirements of eVTOLs (≥ 400 Wh kg⁻¹). Furthermore, maritime shipping ideally requires batteries with energy densities surpassing 1200 Wh L⁻¹, while aviation application like the narrow-body jet requires a minimum battery specific energy of 600 Wh kg⁻¹, emphasizing the significance of developing advanced high-energy-density electrochemical energy storage systems (Fig. 1a)^{2,3}.

Batteries with high specific energy have long been a key area of research. In 2016, the U.S. Department of Energy (DOE) launched the ‘Battery500’ project, aiming to develop batteries with specific energy greater than 500 Wh kg⁻¹⁴. The New Energy and Industrial Technology Development Organization (NEDO) in Japan has set a goal of manufacturing 500 Wh kg⁻¹-level batteries before 2030. In Europe, the proposed “Battery 2030+” roadmap aspires to narrow the gap in specific energy between practical and theoretical energy densities by at least 50%. Meanwhile, China is working to promote the application of batteries with energy densities exceeding 500 Wh kg⁻¹ by 2030, and researchers are now striving to transcend the bar of 600 Wh kg⁻¹. Amongst all rechargeable batteries, lithium metal batteries (LMBs) demonstrate exceptional advantages in delivering high energy densities⁵, especially when paired with the layered ternary oxides with

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high nickel (Ni) contents, moving the specific energy beyond 500 Wh kg⁻¹. Achieving a higher echelon of specific energy, such as 600 Wh kg⁻¹, necessitates more extreme parametric designs (i.e., thick positive electrode, thin negative electrode, and reduced inactive components) and, crucially, the operation of batteries under diminished electrolyte-to-capacity (E/C) ratios^{6,7}, which pose significant challenges for electrochemical stabilities of electrolyte, Li metal negative electrode, and high-voltage positive electrodes. Ether-based electrolytes demonstrate excellent compatibility with the Li metal with significant progress made in this area, such as the development of localized high-concentration electrolytes^{8–10}. Despite the advantages of ether-based electrolytes with imide-based salts, they are intrinsically not oxidation-tolerant enough towards high-Ni-content positive electrode materials under high operating voltages (i.e., LiNi_xCo_yMn_{1-x-y}O₂, $x \geq 0.9$; cutoff voltage ≥ 4.3 V). Besides, the use of imide-based salts causes severe corrosion of aluminum current collectors and sharply increases the cost^{11,12}. Meanwhile, ether electrolytes present poor thermal stability and are prone to serious gassing issues^{5,13–15}, leading to more intractable safety concerns compared to carbonate electrolytes, especially in high-voltage batteries.

Common carbonate electrolytes demonstrate better thermal stability compared to ether electrolytes and offer intrinsic stability towards high-voltage positive electrodes. However, due to their pronounced electron-withdrawing nature, carbonate molecules easily react with the Li metal. Additionally, the solid electrolyte interphase (SEI) formed in the carbonate-based electrolyte usually fails to accommodate the morphological changes of Li, resulting in persistent parasitic reactions between the Li metal and the electrolyte. This leads to Li-inventory consumption, electrolyte depletion, and eventually battery failure^{16,17}. These factors collectively limit the application of carbonate electrolytes in practical-oriented LMBs with lean electrolytes and thin Li foils. Enhancing Li reversibility and reducing electrolyte consumption in carbonate electrolytes could advance LMBs toward broader commercial applications.

Replacing liquid-state carbonate electrolytes with solid-state variants enhances electrode-electrolyte interfacial stability, potentially extending the cycle life of high-energy-density batteries. In recent years, in-situ polymerization technology for gel-solid-state batteries has gained extensive attention due to its compatibility with the existing battery production lines. However, most commercially available ‘semi-solid-state’ batteries are solid-liquid hybrid batteries rather than in-situ polymerization types. In high-energy-density Li-ion batteries, the industrial adoption of in situ polymerization technology has been limited due to unresolved technical challenges, such as poor consistency arising from the uncontrollable polymerization process. Besides, the thermally triggered in-situ polymerization, which is feasible for mass production of gel-solid-state batteries, may lead to unwanted parasitic reactions (e.g. the generation of HF and PF₃) and therefore compromise the performance of the whole battery.

Beyond the realm of electrolyte innovation, preloaded artificial solid-electrolyte interphase (a-SEI) layers present a highly promising avenue for enabling highly reversible lithium plating and stripping processes within carbonate-based electrolytes. The field of a-SEI research has witnessed substantial progress in recent years. Currently, a-SEI designs are primarily categorized into four distinct types: ceramic coatings, polymeric materials, reactive metal salts, and reactive molecules. Each of these categories leverages different material properties and mechanisms to enhance the stability and performance of the lithium-electrolyte interface.

However, the practical deployment of energy-dense LMBs remains stringently constrained by the fabrication complexity and high manufacturing cost. While various electrolyte designs and Li-metal modification strategies continue to emerge, these advances frequently overlook critical practical considerations ranging from associated high costs, and the compromise of Li-metal processability

to the strategy scalability (Supplementary Note 1, Supplementary Fig. 1, Supplementary Table 1). The large-scale manufacturing of energy-dense LMBs demands stricter environmental controls than graphite-based negative electrodes due to the intrinsically more active property of Li metal, requiring lightweight yet effective surface passivation of Li against reactive components in air (moisture, CO₂, etc.) with high cost-efficiency and processability.

In this study, we combine electrolyte engineering with efficient Li metal surface modification to enable the scalable fabrication and stable operation of an Ah-class pouch-type high-energy-density LMB under lean carbonate electrolyte conditions. The in-situ constructed polymer backbone networks effectively confine carbonate solvents and mitigate electrolyte decomposition. Meanwhile, the formulated bicontinuous gradient polymer layer (BGPL) on Li not only enhances the reaction kinetics of the Li metal but also protects fresh Li from detrimental moisture, oxygen, and invasive electrolyte components. Specifically, the BGPL@Li enables exceptional reversibility of Li plating/stripping, with a Coulombic efficiency (CE) of up to 99.66% in carbonate electrolytes. The cell, comprising a Ni92 (LiNi_{0.92}Co_{0.06}Mn_{0.02}O₂) positive electrode, in-situ gelatinous electrolyte, and BGPL@Li negative electrode (50 μm; ratio of the negative electrode capacity to the positive electrode capacity, i.e., N/P ratio, of 3.1), achieves a high capacity retention of 81.23% after 405 cycles, with an average CE of 99.94%. A high specific energy of 604.2 Wh kg⁻¹ (1153 Wh L⁻¹) is obtained in the 11 Ah-class pouch cell, with a lifespan of over 100 cycles and an energy retention of 92.83% at the 100th cycle. This research amalgamates a highly scalable Li modification strategy with gel-solid-state electrolyte engineering, establishing an inspiring and applicable paradigm to overcome the specific energy limitations of long-cycle-life LMBs.

Results and discussion

Parametric designs for 600 Wh kg⁻¹ batteries

Figure 1b illustrates the multistep transition of cell-level specific energy from state-of-the-art commercial batteries with the graphite/carbon-coated silicon (Gr/Si@C) negative electrode towards LMBs with high energy densities (design parameters of baseline Gr/Si@C||NCM811 cell and other cells shown in Supplementary Table 3). This trend demonstrates that achieving high specific energy requires breakthroughs both in battery materials and the extreme parametric design of the battery. Figure 1c specifically depicts the evolution in the specific energy of pouch cells based on diverse negative electrode-positive electrode pairs, especially when varied E/C ratios are applied. To enable an high specific energy of 600 Wh kg⁻¹, preferably, thick high-capacity positive electrodes (specific capacity ≥ 220 mAh g⁻¹; areal capacity ≥ 5.5 mAh cm⁻²) should be incorporated into the LMB design. As discussed in Supplementary Note 2, generally, commercialized LiNi_{0.8}Co_{0.1}Mn_{0.1}O₂ (NCM811) positive electrodes typically deliver specific capacities of around 200 mAh g⁻¹ or lower (195 mAh g⁻¹ assumed for comparison in this study), while the high-Ni-content positive electrode, LiNi_xCo_yMn_{1-x-y}O₂ ($x \geq 0.9$), can meet the requirement of specific capacity ≥ 220 mAh g⁻¹. Regarding the negative electrode, replacing the Gr/Si@C composite with the Li metal promotes the battery specific energy (the blue vs. orange region in Fig. 1c). As for the positive electrode, LMBs with Ni92 positive electrodes exhibit disparately higher energy than NCM811 with the same pouch cell design (the purple vs. blue region in Fig. 1c). Furthermore, the proportions of non-active components such as the current collector, separator, and packaging materials should be minimized to the greatest extent feasible. Additionally, the E/C ratio should be lower than 1.0 g Ah⁻¹ to surpass the 600 Wh kg⁻¹ threshold, which is impractical for cells with thick yet porous negative electrodes but achievable for the LMB with the planar Li metal. Multiple strategies, ranging from electrolyte engineering to Li metal modification should be concurrently integrated into the battery fabrication for achieving long-cycle-life, energy-dense LMBs under lean electrolyte conditions (Fig. 1d)¹⁸.

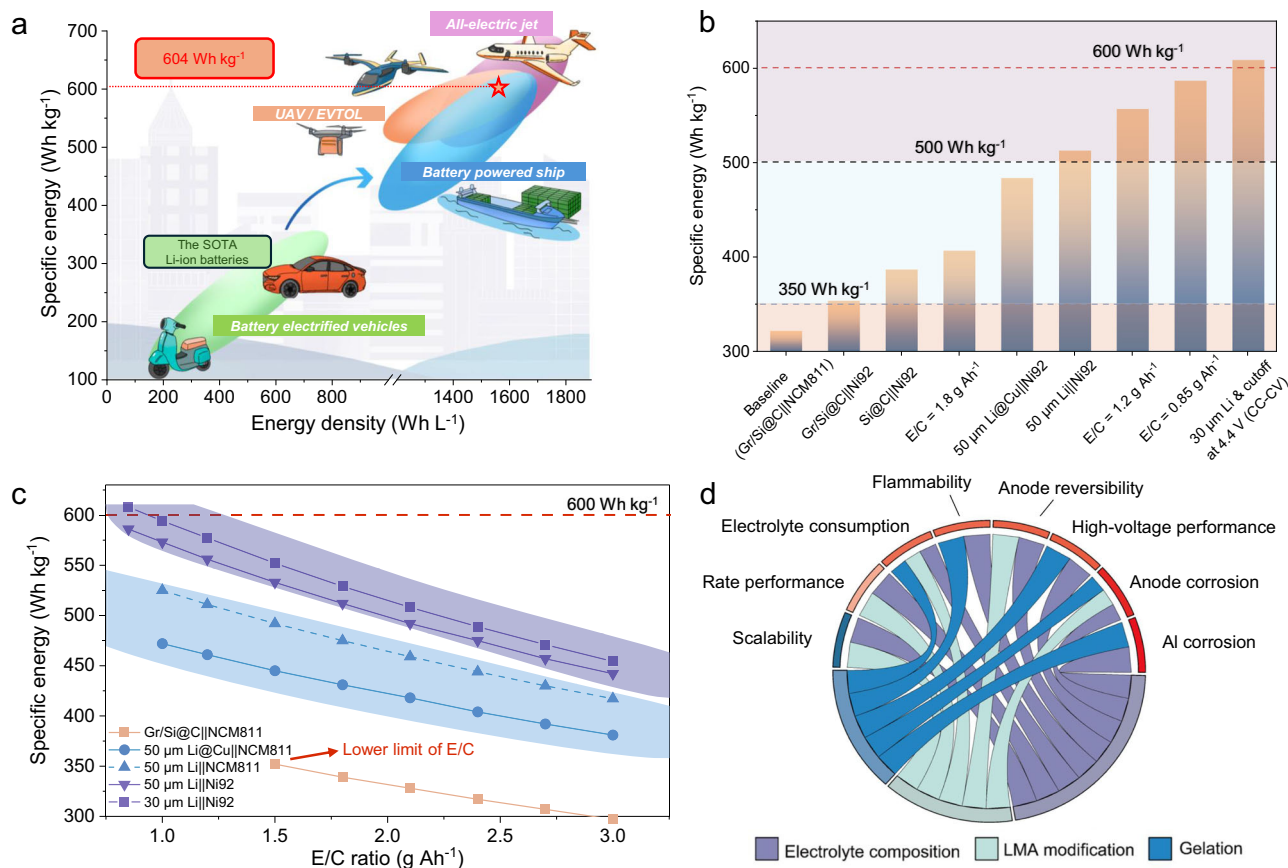


Fig. 1 | Significance and future pathways of the 600 Wh kg^{-1} LMB. a The evolution of secondary Li-based batteries and their future applications. **b** Calculation of the specific energy of pouch cells under different cell design parameters. The E/C ratio here represents the preserved amount, which is lower than the injection doses. **c** Cell-level energy densities of batteries as a function of E/C ratio for various

negative electrode-positive electrode couples. Details of these cells are presented in Supplementary Table 3 and further discussed in Supplementary Note 2. **d** Chord graph illustrating three main factors affecting the performance and production scalability of LMBs.

In-situ electrolyte gelation and performance analysis

The carbonate-based MFN electrolyte used in this work is first optimized to enhance Li reversibility (MFN: 1.2 M LiPF_6 in FEC/FEMC/EMC/DMC ($v/v/v/v = 3:1.5:1.5:4$) + 0.02 M $\text{Mg}(\text{TFSI})_2$ + 65 mM LiNO_3 + 0.1 wt % hexamethyldisilazane (HMDS), M: denoted as $\text{Mg}(\text{TFSI})_2$; F: denoted as FEMC; N: denoted as LiNO_3) (Supplementary Fig. 2). The MFN electrolyte achieves a high lithium CE of 99.16% (Li||Cu cell test) by forming a LiF-rich SEI on the Li metal and sustains high voltages up to 4.7 V (Li||Ni92 floating analysis), outperforming the conventional carbonate electrolyte, LP334 (LP334: 1 M LiPF_6 in EC/DEC/DMC ($v/v/v = 3:3:4$) + 2 vol% FEC + 2 vol% VC) (Supplementary Note 3, Supplementary Figs. 3–14). The incorporation of LiNO_3 not only enhances Li reversibility, a well-established effect in liquid electrolytes but more importantly, it prevents uncontrolled polymerization of the gel-solid-state electrolyte precursor solution in this work. The controllability of the gelation process helps promote the consistency of as-assembled gel-solid-state batteries, which is essential for large-scale commercial deployment of solid-state batteries but has been overlooked in previous relevant battery research. Meanwhile, HMDS, a low-cost yet critical molecular additive, scavenges HF in the electrolyte, which is readily generated during the high-temperature incubation of the in-situ polymerization process and can seriously degrade the fluoroethylene carbonate (FEC)-containing LiPF_6 -based MFN electrolyte.

To further enhance the positive electrode stability and reduce the anodic side reactions, anchoring the MFN electrolyte through establishing an interconnected polymer framework represents a highly efficacious strategy for further enhancing the stability of positive

electrode^{19,20}. Herein, three functional monomers including pentaerythritol tetraacrylate (PETEA), diethyl allyl phosphate (DAP), and 2,2,3,4,4,4-hexafluorobutyl methacrylate (HFBMA) are molecularly “spliced” together to construct a crosslinked polymer backbone through a facile in-situ free radical polymerization to confining the MFN (GMFN; G: denoted as gelatinous). PETEA with the four-armed C=C double bonds enables easy cross-linking between different monomers at a low mass fraction. DAP with single-arm allyl phosphates can help dissociate LiNO_3 and guarantee fast Li^+ transport while copolymerizing with other monomers. The massive $\text{C}^{\delta+}\text{F}^{\delta-}$ bonds in HFBMA have higher local dipole moments compared to the C=O bond due to the electronegativity distinction, which induces intermolecular interaction. The F- and P-donating polymer backbones formed in the GMFN electrolyte can bind with solvents and anions, alleviating electrolyte consumption. Additionally, these networks facilitate the formation of both the F-rich solid electrolyte interphase (SEI) and the P-rich cathode-electrolyte interphase (CEI), and bolster the flame-retardant properties of electrolytes by capturing $\text{H}^{\cdot}$ radicals (Fig. 2a)^{21,22}.

Molecular dynamics (MD) simulations are carried out to investigate the molecular anchoring behavior of the crosslinked polymer backbone toward carbonate solvents (Supplementary Figs. 15a and 16). The polymer backbone, composed of PETEA, DAP, and HFBMA, exhibits enhanced binding energies towards solvent molecules (Fig. 2b, c), exceeding the interactions among solvent molecules, which suggests the migration of solvents can be restricted. The stronger binding energies between the polymer skeleton and solvent molecules mainly

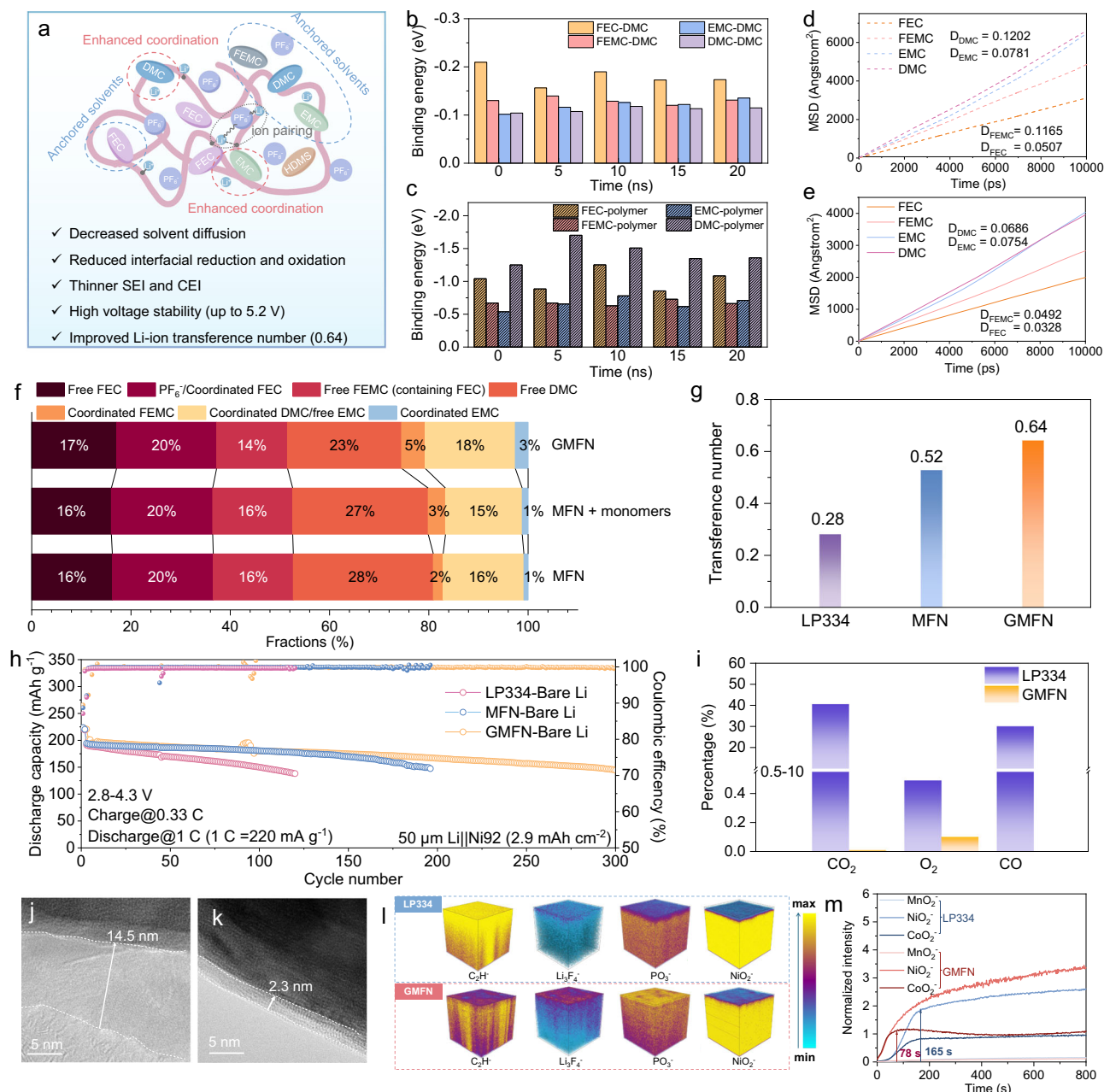


Fig. 2 | Schematics, simulations, and characterizations of the GMFN electrolyte.

a Schematic presentation of how the GMFN functions and the as-brought merits (only the major salt, LiPF_6 , is presented here). **b, c** Binding energies of the solvent molecules to each other (**b**) and to the polymer backbone (**c**) consisting of PETEA, DAP, and HFBMA. **d, e** Diffusion coefficients of solvent molecules in the MFN (**d**) and GMFN (**e**) electrolytes. **f** Fractions of the corresponding peak area derived from Raman spectra of the MFN electrolyte. MFN + monomers (MFN + monomers: denoted as the precursor solution of the GMFN electrolyte before the polymerization process), and GMFN electrolyte. **g** Li^+ transference number of the LP334, MFN, and GMFN electrolytes. **h** Cycling performance of the $50\ \mu\text{m}\ \text{Li}||\text{Ni92}$ cells

based on LP334, MFN and GMFN electrolytes (25°C). **i** Contents of CO_2 , CO , and O_2 in the generated gas based on GC-MS results of the LP334- and GMFN-cycled 5 Ah-pouch cells. These two pouch cells were cycled at 25°C to their end-of-life (80% capacity retention) and then tested in a fully discharged state. **j, k** TEM images of the LP334-cycled (**j**) and GMFN-cycled (**k**) Ni92 particles. **l** ToF-SIMS 3D rendered images of the C_2H^+ , Li_3F_4^+ , PO_3^- , and NiO_2^- . **m** Normalized intensity curves of the NiO_2^- , MnO_2^- and CoO_2^- as the function of etching time. These particles and electrodes for both TEM and ToF-SIMS characterizations are collected from $50\ \mu\text{m}\ \text{Li}||\text{Ni92}$ cells after 30 full charge/discharge cycles at $0.33\text{C}/1\text{C}$ (25°C).

originate from the multisite dipole-dipole interaction between the $\text{C}=\text{O}$ bonds of solvent molecules and polar $\text{C}-\text{F}$ bonds of the HFBMA and $\text{P}=\text{O}/\text{P}-\text{O}$ bonds of the DAP (Supplementary Fig. 15b)^{22–24}. Diffusion coefficients of various solvent molecules in the MFN and GMFN electrolytes were computed based on the MD simulation results (Fig. 2d, e). The solvent molecular diffusion is substantially retarded after the electrolyte gelation. Raman spectroscopy was utilized to investigate the solvation structure evolution triggered by the polymer-solvent

interaction. Figure 2f shows the statistical ratios of FEC/DMC/EMC/FEMC molecules at both free and coordinated states, which are derived from the Raman results (700 cm^{-1} – 960 cm^{-1}) (Supplementary Figs. 17 and 18)^{20,25,26}. The results reveal that the free non-fluorinated carbonate molecules including DMC (915 cm^{-1}) and EMC (933 cm^{-1}) decrease after polymerization. The decrease of free non-fluorinated carbonates and their increased coordination with Li^+ contribute to enhancing the oxidation resistance of the electrolyte system^{20,27}. The

desolvated free-state FEC solvent has a higher lowest unoccupied molecular orbital (LUMO) compared to its coordinated state, which reduces its likelihood of reacting with Li metal and becoming overconsumed²⁸. However, Raman results roughly illustrate states of solvent molecules and ions but cannot explicitly describe how the solvent coordination changes and how the polymer skeleton interacts with the solvents.

To validate the change in the solvation structure, the wide-angle X-ray scattering (WAXS) of the MFN and GMFN electrolytes was performed, as detailed in Supplementary Note 4 (Supplementary Fig. 19). These WAXS results indicate an enhanced interaction between Li^+ and PF_6^- , consistent with the findings from the RDF results.

To further probe this intermolecular interaction between the polymer backbone and solvents, ^1H , ^7Li , ^{19}F , ^7Li , and the ^1H - ^{19}F heteronuclear Overhauser effect spectroscopy (HOESY) nuclear magnetic resonance (NMR) analyses were conducted (Supplementary Figs. 20–24). The corresponding results elaborated in Supplementary Note 4 strongly corroborate our assumption that the polymer backbone effectively anchors the solvent molecules (Supplementary Fig. 25). Besides, the differential scanning calorimetry (DSC) analysis (Supplementary Fig. 26) indicates that the ternary PETEA-HFBMA-DAP skeleton effectively retard the solvent evaporation. This thermal stabilization provides direct experimental validation of the strong polymer-solvent binding interaction. Radial distribution functions (RDFs) demonstrate a distinction in the solvation structure of Li^+ in the MFN and GMFN electrolytes. The increase in the coordination number (CN) of PF_6^- to Li^+ from 0.18 in the MFN to 0.22 in the GMFN probably suggests that more ion-pair configurations exist in the GMFN electrolyte compared to the liquid MFN electrolyte (Supplementary Fig. 27; Supplementary Note 4), which is beneficial in forming the thin and robust SEI. Furthermore, the CN of FEC to Li^+ decreases by 0.27 after gelation, indicating a higher prevalence of free FEC within the GMFN electrolyte.

The in-depth X-ray photoemission spectroscopy (XPS) was performed on the cycled Li metal in the GMFN electrolyte (with the morphology shown in Supplementary Fig. 28). The Li metal cycled in the GMFN electrolyte shows a lower overall intensity of C 1s signal compared to the corresponding spectra of MFN and LP334 (Supplementary Figs. 11 and 29). Moreover, the Li metal cycled in the LP334 and MFN electrolytes show no evidence of Li^0 (53.2 eV), while the Li metal cycled in the GMFN electrolyte shows a crescendo signal of Li^0 as the etching time increases from 0 s to 600 s (Supplementary Fig. 30). Therefore, it is reasonable to conclude that the GMFN electrolyte promotes the formation of a thinner, inorganic-rich SEI on the Li metal, likely due to suppressed solvent reduction and increased ion pairs. Additionally, the conductivity (at 25 °C) of the solid-state GMFN electrolyte is measured to be 1.82 mS cm^{-1} (Supplementary Fig. 31). Notably, the t_{Li^+} of the GMFN electrolyte reaches 0.64, surpassing that of the commercially available LP334 (0.28) and the MFN (0.52) electrolytes (Fig. 2g and Supplementary Fig. 32).

The enhanced cathodic stability in the GMFN electrolyte is examined by the floating test. The $\text{Li}||\text{Ni92}$ cell with the GMFN shows a much weaker leakage current compared to that with the MFN electrolyte, particularly at 4.7 V or above (Supplementary Figs. 33 and 34). The leakage current of the $\text{Li}||\text{Ni92}$ cell using the GMFN electrolyte remains stable even at 5.2 V, confirming that the in-situ gelation process promotes the oxidative stability of the electrolyte. The cycling performance of the GMFN electrolyte is evaluated using 50 μm $\text{Li}||\text{Ni92}$ cells in a 0.33 C/1 C charge/discharge test (Fig. 2h). The cell with the GMFN electrolyte shows a capacity retention of 82.64% (from 201.6 to 166.6 mAh g^{-1}) after 200 cycles, outperforming the LP334 (72.36% after 120 cycles) and MFN (75.82% after 196 cycles) electrolytes. The electrochemical impedance spectroscopy (EIS) of 50 μm $\text{Li}||\text{Ni92}$ cells at different state of charges (SOCs) shows that the cell with the GMFN shows a decreasing trend of impedance from the 1st to 10th cycle

(Supplementary Fig. 35). It is found that the cell using the GMFN electrolyte shows larger impedances (charge transfer resistance, R_{ct} , and interface resistance, $R_{\text{interface}}$) When the cell is 50% charged (SOC = 50%), the R_{ct} and $R_{\text{interface}}$ of the GMFN electrolyte are decreased and show little difference from that of the MFN electrolyte. Furthermore, when the cell is fully charged (SOC = 100%), the R_{ct} and $R_{\text{interface}}$ of the GMFN electrolyte are lower than that of the MFN electrolyte. This EIS result suggests that interfaces built in GMFN are more stable compared to the MFN electrolytes^{29,30}.

Furthermore, the gas chromatography-mass spectrometry (GC-MS) characterization was carried out on 5 Ah-class pouch cells to assess the stability of positive electrode in terms of gas generation in the electrolyte (Fig. 2i and Supplementary Figs. 36 and 37). Gases generated in the 5 Ah-class pouch cells with the LP334 electrolyte (parametric information in Supplementary Table 4) contain significant amounts of CO_2 (40.54 %) and CO (30.04%), along with an ignorable amount of O_2 (0.41%). In contrast, these gases are almost undetectable in the GMFN-based pouch cell, illustrating that the GMFN electrolyte provides an enhanced intrinsic oxidation resistance and stability against layered oxide materials with high Ni content.

A thin yet dense CEI not only mitigates transition metal dissolution but also boosts ion diffusion, serving as a key indicator of achieving both a stable positive electrode and a stable electrolyte³¹. Transmission electronic microscope (TEM) images of positive electrode particles undergoing 100 cycles in LP334 and GMFN electrolytes are shown in Fig. 2j, k. The Ni92 particle cycled in the LP334 electrolyte shows a nonuniform CEI with a thickness of 14.5 nm, far exceeding the CEI formed in the GMFN (2.3 nm). TOF-SIMS characterization was performed to visualize the element and component distributions of CEI. The CEI formed in the LP334 electrolyte shows a higher intensity of C_2H^+ and C_2HO^+ while a lower intensity of PO_3^- and $\text{Li}_x\text{F}_{x+1}^-$ segments (Fig. 2l and Supplementary Figs. 38 and 39), implying that the CEI of Ni92 particles formed in the LP334 is more abundant in organic compounds with fewer inorganic fluoride components like LiF. The CoO_2^- and NiO_2^- signals of cycled Ni92 particles in the LP334 electrolyte are undetectable initially (Supplementary Fig. 40). The CoO_2^- signal intensity of the cycled Ni92 particles reaches steady after 165 s, which is longer than the 78 s of the cycled Ni92 particles in the GMFN, indicating that the CEI of the cycled Ni92 particles in the LP334 electrolyte is thicker (Fig. 2m). The higher thickness of CEI rendered by the LP334 electrolyte mainly ascribes to continuous parasitic reactions occurring between the electrolyte and positive electrode material. The above results are consistent with CEI thicknesses observed in TEM images, which show that the GMFN electrolyte forms a thinner CEI with more stable components. This contributes to the improved stability of the positive electrode, the electrolyte, and ultimately the full cell.

More attractively, the solid-state GMFN electrolyte manifests good flame-retardancy (Supplementary Fig. 41). The conventional LP334 electrolyte is highly flammable and continues to burn until the electrolyte is fully consumed. The liquid MFN and “MFN+monomers” electrolytes are also flammable, with the flame lasting for over 5 seconds. In contrast, the solid-state GMFN electrolyte cannot be ignited even when continuously heated with an ignitor for over 5 seconds. This suggests that the crosslinked polymer network imparts enhanced flame-retardancy to the solid-state GMFN electrolyte, potentially improving battery safety performance.

Design and properties of the BGPL@Li

While the gelation strategy enhances the cathodic stability of electrolytes, inadequate interfacial contact between gel-solid-state electrolytes and the Li metal often induces non-uniform lithium plating. Consequently, most thermally initiated in-situ polymerized electrolytes fail to achieve highly reversible lithium plating/stripping behavior ($\text{Li}||\text{Cu}$ cell efficiency <99.0%), a shortfall that renders them unsuitable for sustaining long-term operation in practical high-areal-capacity

LMBs. In addition to electrolyte engineering, strategies for constructing artificial SEIs (a-SEI) have attracted significant attention for improving the reversibility of Li^{31–35}. Coating lithium (Li)-metal with a rationally designed interfacial polymer layer homogenizes lithium-ion flux at the Li-metal/electrolyte interface, effectively resolving the issue of uneven lithium distribution at the gel-solid-state electrolyte-Li interface. Simultaneously, a dense polymer layer can mitigate crosstalk from transition metal ions leached from the positive electrode³⁵, as well as parasitic reactions on Li caused by residual unpolymerized monomers (containing reactive C=C bonds) or leftover initiators (i.e., BPO). Therefore, synergizing gel-solid-state electrolytes with polymer a-SEI may simultaneously address persistent challenges of Li-metal, high-voltage positive electrodes, and electrolyte depletion.

Beyond the reversibility-related issues, the manufacturing of the LMB is significantly hindered by the poor processability of soft, sticky, and moisture-/air-sensitive Li foils during cell fabrication, which is generally overlooked³². A viable, multifunctional a-SEI constructing strategy for practical applications must address all of these aspects simultaneously. Ideally, the a-SEI, derived from inexpensive and low-toxicity chemicals, should provide highly conductive pathways for Li-ions, thereby reducing interfacial and charge transfer resistance. It should also form a uniform coating on the Li metal, effectively blocking exposure to air and preventing corrosion from electrolyte components³³. Moreover, the preparation of the a-SEI should be compatible with pilot-scale processes such as roll-to-roll tape casting and spray coating for mass production. Additionally, the a-SEI should impart rigidity to the Li foils, preventing the glutinous, easily strained Li from undergoing irreversible deformations and enabling a damaging-free cell assembly process. This property also helps avoid morphological inhomogeneity and uneven stress distribution in the pouch cell, thereby ensuring a more efficient and reliable cell assembly process.

A solution-processing method is developed to enable the scalable tape-casting technique for preparing the BGPL-modified Li foil (BGPL@Li). 1,3,5-Trioxane (TX), though previously reported in electrolyte systems, has rarely been exploited for the Li metal modification. TX is selected for its nonvolatile (solid-state) property. It autonomously polymerizes on the Li metal (Supplementary Fig. 42), forming a polyether-segment-rich layer, poly-TX (PTX), on the surface of Li metal (Supplementary Note 1 Part II). The polyether-based layer has abundant ether-oxygen sites and low interface energy with the Li metal, enabling efficient Li-ion transport and strong adhesion to the Li metal³⁴. However, the a-SEI derived from the TX alone is inhomogeneous, which limits its effectiveness in providing consistent protection (Supplementary Fig. 43). In contrast, the hydrophobic PVDF-co-HFP promises excellent film-forming properties, which provides superior moisture and air resistance. Additionally, its high dielectric constant (ϵ_r) contributes to a high exchange current density. However, the PVDF-co-HFP possesses a high interface energy with Li metal and thus leads to poor Li-affinity. The as-coated PVDF-co-HFP layer is easily peeled off from the Li metal surface. Therefore, combining the TX and PVDF-co-HFP to construct a homogeneous, hydrophobic, and lithophilic a-SEI with a bicontinuous gradient structure can effectively address the above issues (Fig. 3a, d). Furthermore, a small amount of lithium difluoro(oxalato)borate (LiDFOB) is incorporated into the precursor solution, which enhances the Li-ion conductivity of the a-SEI³⁵.

The cross-sectional scanning electron microscope (SEM) image reveals the uniform BGPL is around 2.1 μm thick and seamlessly coalesces with the Li metal without any gap (Fig. 3b, c). Such an a-SEI has a bicontinuous gradient distribution of the PTX and PVDF-co-HFP as revealed by the XPS in-depth profiles (Supplementary Fig. 44). The BGPL endows Li foils with multidimensional appealing properties. The BGPL keeps intact and remains non-swelled when soaked in these carbonate electrolytes for 24 h and withstands high temperatures up to 70 °C (Supplementary Figs. 45 and 46). Raman spectra of the freshly

prepared BGPL@Li and the electrochemically stripped BGPL from Li foils in MFN electrolyte are shown in Supplementary Fig. 47. The main characteristic peaks of the BGPL remain intact after electrochemical stripping, further demonstrating the stability of the BGPL in the carbonate electrolyte environment.

Notably, for use in practical stacking-layer cells, the soft and sticky Li foil must be appropriately pretreated to prevent deformation and adhesion to the metallic components of the stacking devices. The issues of wrinkling and warping in the Li (or Li@Cu) foils must be addressed to ensure morphological flatness and homogeneous stress distribution in pouch cells during cycling. The excellent mechanical properties of the BGPL@Li eliminate the problem of Li-foil wrinkling, which leads to the inhomogeneity of the electrodes and stacked pouch cells (Fig. 3e, Supplementary Figs. 48 and 49). Furthermore, for scale-up use and long-term storage, the modified Li foil should be oxygen- and moisture-resistant to ensure the lengthy assembly process of the Li foil in a dry room and long-term storage in dry air, where the dew point is generally higher than -40 °C.

The BGPL@Li shows superior moisture- and oxygen-tolerance. The BGPL@Li remains stable when exposed to water spills, while uncoated Li foil reacts vigorously and continuously generates hydrogen upon contact with water. This highlights the good hydrophobic and airtight properties of BGPL (Fig. 3f, Supplementary Fig. 50, and Supplementary Video 1). In addition, the double-side coated BGPL@Li remains fresh after being sealed in a container filled with dry air (dew point of -30 °C) for over 240 h, with more than 96% of the active Li preserved. In contrast, the bare Li foil turns black after 96 h (Supplementary Figs. 51 and 52). The mossy oxidized area (LiOH) of the bare Li keeps expanding and develops noticeable cracks within 240 h. Supplementary Fig. 53 shows the surface evolution of the Li foil coated by different layers exposed to the moist atmosphere (relative humidity (R.H.) = 60%). The entire TX@Li and the edges of PVDF-co-HFP@Li turn black upon exposure to air, while the BGPL@Li remains intact without any signs of corrosion, suggesting its excellent moisture and oxygen tolerance.

Furthermore, the BGPL promotes the exchange current density of Li foil (Fig. 3g and Supplementary Fig. 54). The symmetric cell of BGPL@Li delivers the lower electrolyte/electrode resistance (R_{EE}) and charge transfer resistance (R_{ct}) compared with bare Li. The improved exchange current density and reduced impedance originate from both the high ϵ_r of PVDF-co-HFP and Li-ion conducting lithophilic PTX (the PVDF-co-HFP coating alone leads to unsatisfying performances of batteries, Supplementary Note 5, Supplementary Fig. 55). The low cost, low toxicity, and excellent film-forming properties of the coating lay the basis for the BGPL@Li to be prepared on a pilot scale for mass production, as presented in Fig. 3h. Piles of energy-dense LMBs were easily prepared with BGPL@Li foils with the nontoxic and Li-benign solvent (Supplementary Figs. 56–58, Supplementary Table 2) using automatic equipment (Supplementary Fig. 59), formulated into battery modules and successfully powering our customized UAV (Supplementary Video 2).

Electrochemical performance of the BGPL@Li in solid-state full cells

After elucidating the preparation and advantages of BGPL@Li for large-scale production of LMBs, we proceed to investigate the electrochemical performance of BGPL@Li. As shown in Fig. 4a, Li||Cu asymmetric cells are used for the characterization of Li reversibility. The asymmetric cell using the BGPL-coated Cu foil delivers a high CE of 99.66%, even when tested with the carbonate-based MFN electrolyte, representing one of the highest reported values in carbonate electrolyte systems (Fig. 4b)^{16,36–43}. With the solid-state GMFN electrolyte, the BGPL@Li still delivers a high CE of 99.42% (with the overpotential information shown in Supplementary Fig. 60), which is one of the highest Li efficiencies for gel-solid-state electrolytes

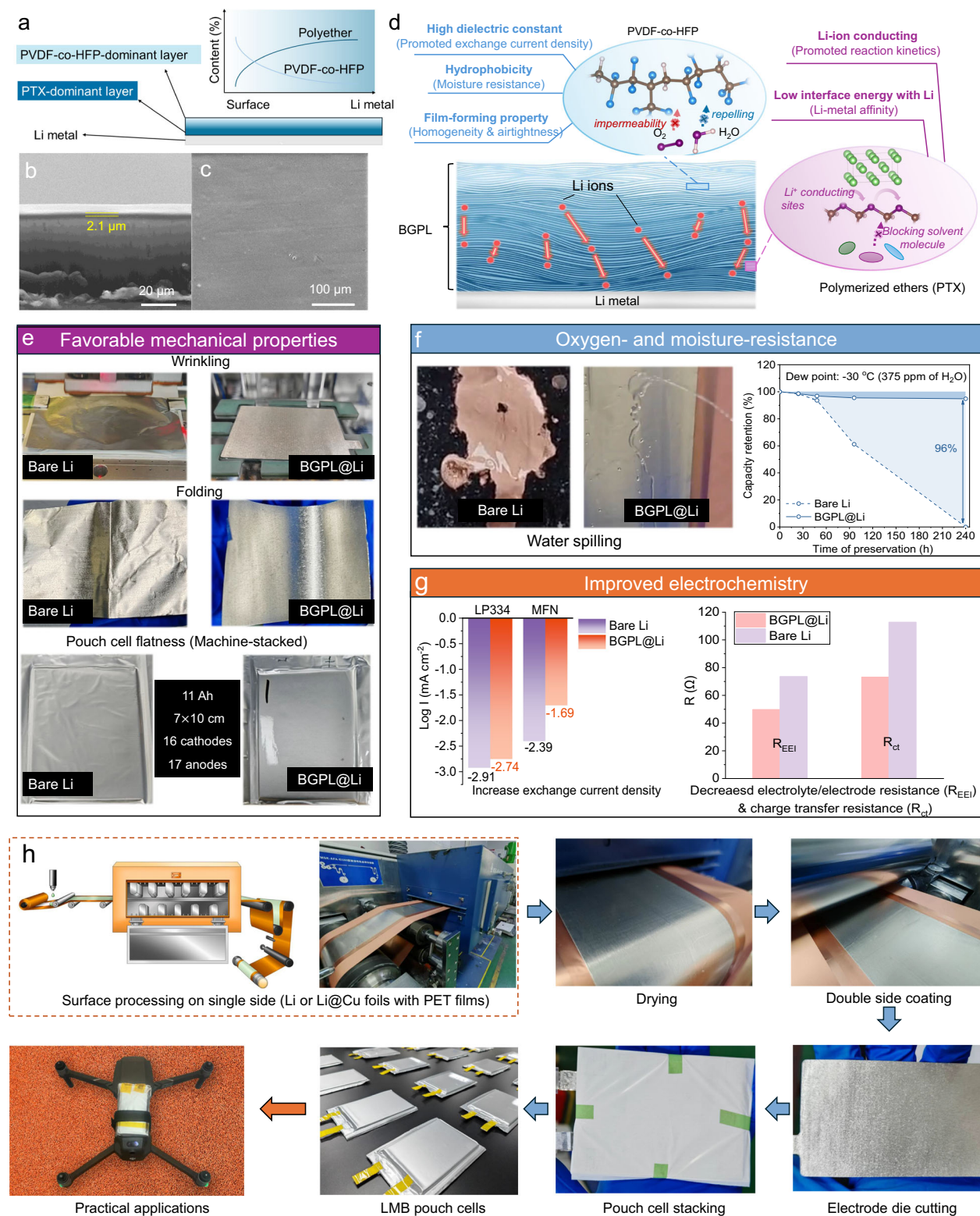


Fig. 3 | Schematics, manufacturing, and properties of the BGPL@Li. **a** Structural schematics of the BGPL@Li. **b**, **c** SEM images from the cross-sectional view (**b**) and top view (**c**). **d** Schematic presentation delineating the functions of each component in BGPL@Li. **e** Showcases of the improved mechanical properties for the prevention of wrinkling and folding, resulting in high-quality cell assembly.

f Moisture- and oxygen-proof properties of BGPL@Li foils. **g** Electrochemical performance improvement of the BGPL@Li compared to bare Li based on symmetric cell (with an injected electrolyte of 30 μL) tests in terms of Tafel and Nyquist plots. **h** Schematic workflow showing the scalable preparation of BGPL@Li foils, BGPL@Li assembled pouch cells, and their applications in UAV.

(Supplementary Table 5). The SEM image of the plated Li (top view) is shown in Fig. 4c. The plated Li using the BGPL@Li with the MFN electrolyte exhibits a densely packed morphology with large granular sizes (up to 40 μm). Meanwhile, the SEM image of the FIB-processed

cross-section shows the dense morphology of the BGPL@Li in both the MFN and GMFN electrolyte (Fig. 4d; Supplementary Fig. 61), in sharp contrast to the porous structure of the plated Li observed with LP334 and MFN electrolytes on bare Cu foil. This indicates that the BGPL@Li

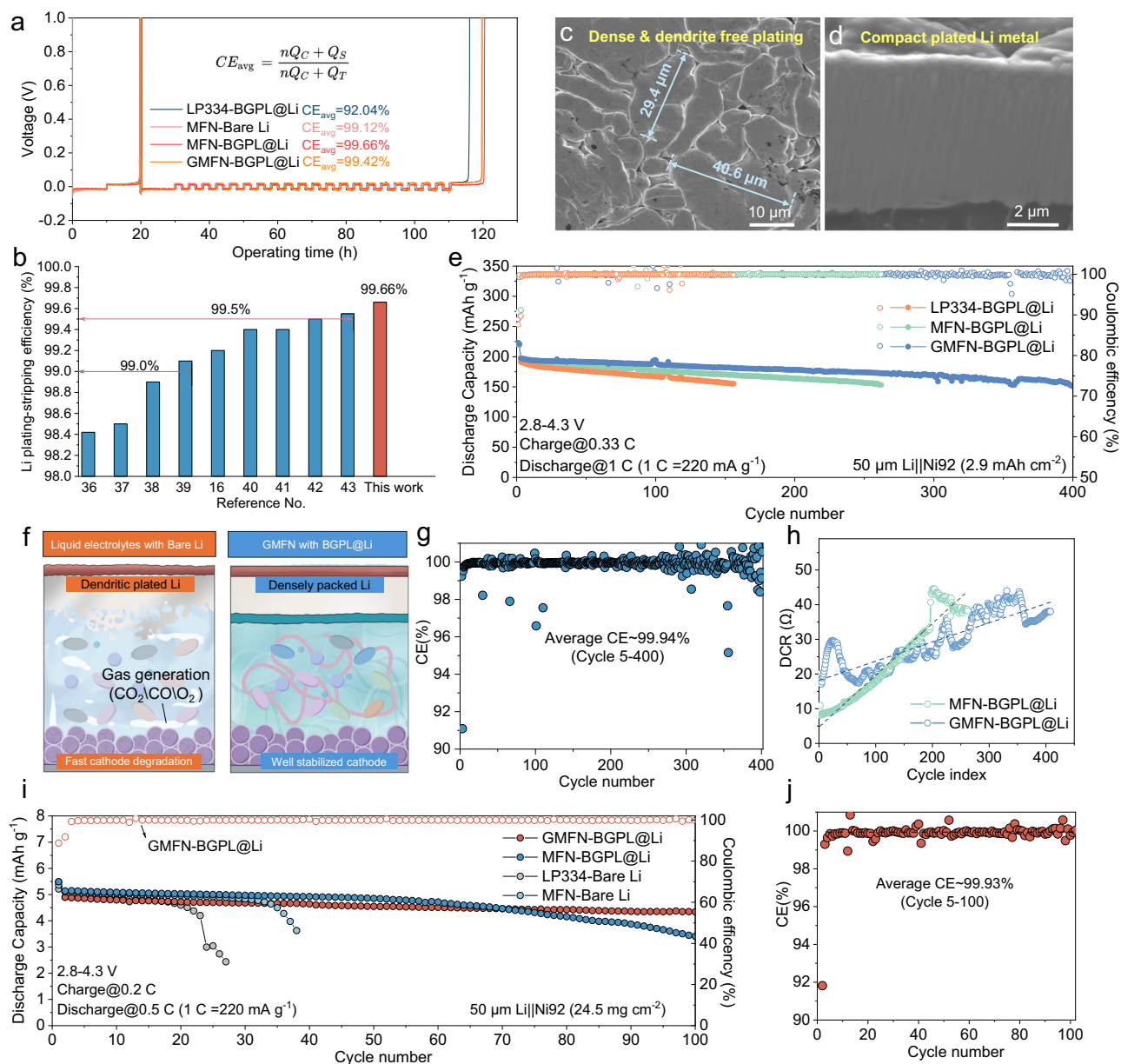


Fig. 4 | Electrochemical performance of the BGPL@Li and the application in gel-solid-state full cells. a Plating/stripping efficiency tests by the Aurbach method of the BGPL@Li with diverse electrolytes. **b** Comparison of reported Li plating/stripping efficiencies to this work. **c, d** SEM images of plated Li (0.5 mA cm⁻², 1.0 mAh cm⁻²) with the BGPL@Li foil and the MFN electrolyte from the top view (**c**) and cross-sectional view (FIB-processed) (**d**) after 20 plating-stripping cycles at 0.5 mA cm⁻² and 1.0 mAh cm⁻². **e** Cycling of BGPL@Li||Ni92 cells using LP334/MFN/GMFN electrolytes at 0.33 C/1 C (25 °C, with 2 activation cycles at

0.1 C/0.1 C charge/discharge). **f** Schematics illustrating the coordination between BGPL@Li and GMFN electrolyte enabling a dendrite-free LMB with an enhanced positive electrode stability. **g** CEs of the BGPL@Li||Ni92 cell with the GMFN electrolyte. **h** DCR of the BGPL@Li||Ni92 coin cells using the MFN and GMFN electrolytes (along with linear fitting curves). **i** Cycling of cells using high-loading Ni92 positive electrodes with different negative electrodes or electrolytes. **j** CEs of the BGPL@Li||Ni92 cell with the GMFN electrolyte.

is highly effective in regulating the Li plating behavior, leading to the formation of homogeneous, densely packed, and dendrite-free Li.

For the effective operation of a full LMB, both a highly stable positive electrode and reversible Li plating/stripping are essential. Since the solid-state GMFN electrolyte is effective in stabilizing high-voltage, high-Ni positive electrode materials, coupling the Li-protective and Li-regulating BGPL with the solid-state GMFN electrolyte could enhance the full-cell performance. Figure 4e shows the cyclability of the assembled full cells based on the BGPL@Li with various electrolytes. The BGPL@Li||Ni92 cell with the solid-state GMFN electrolyte demonstrates a high capacity retention of 81.23% after 405 cycles under a 0.33 C/1 C charge/discharge cycle mode

(Supplementary Fig. 62), surpassing the retention rate of the cells with the LP334 and MFN electrolytes (LP334: 81.21% after 155th cycle; MFN: 77.59% after 260th cycle). The slower decay rate of the BGPL@Li||Ni92 full cell with the solid-state GMFN electrolyte can be attributed to the combined effects of improved Li metal reversibility, enhanced positive electrode stability, and minimized degradation of the electrolyte. (Fig. 4f). Moreover, the average CE of the full cell with the solid-state GMFN electrolyte reaches 99.94%, outperforming all other tested cells (Fig. 4g and Supplementary Fig. 63). This underscores that enhanced stability of the electrolyte/electrode interfaces in the full cell can be achieved through the synergistic combination of electrolyte engineering (GMFN) and Li modification (BGPL@Li).

The evolution of direct current resistance (DCR) offers another glimpse of the interfacial stability within full cells (Fig. 4h). The BGPL@Li||Ni92 cell with the GMFN electrolyte shows an initial DCR of 20.06 Ω , which is higher than that with the MFN electrolyte (10.95 Ω), and gradually increases within the first 25 cycles. However, it subsequently decreases to 17.39 Ω at the 70th cycle, with an overall slower increase compared to the cell with the MFN electrolyte. After 180 cycles, the DCR of the cell with the MFN electrolyte surpasses that of the cell with the GMFN, rising to 44.73 Ω at the 298th cycle, while the DCR of the cell with the GMFN remains at a lower 38.08 Ω at the 407th cycle. The DCR results indicate that the solid-state GMFN electrolyte contributes to more stable interfaces, with less resistance growth over the long-term cycling of the full cell.

Coin-type full cells were assembled and cycled with the high-loading Ni92 electrode (5.5 mAh cm⁻²) (Fig. 4i). The cell with the GMFN delivers an initial activation capacity of 5.47 mAh cm⁻² under a 0.1 C/0.1 C charge/discharge rate, which is slightly lower than that of the cell with the MFN electrolyte (5.49 mAh cm⁻²). After the applied current increases to 0.2 C/0.5 C, the cell with the GMFN exhibits slightly lower capacity utilization compared to those with liquid electrolytes. This is likely due to insufficient contact between the solid-state gel-polymer network of GMFN and the thick calendared electrode (~78 μ m) in the assembled coin cells, resulting from the absence of a vacuum standing process—a step commonly employed in the preparation of pouch cells. However, the full cell with the GMFN enables stable cycling and maintains 88.84% of its capacity after 100 cycles at 0.2 C/0.5 C, far higher than the 65.3% for the cell with the MFN electrolyte. Especially, the average CE reaches 99.93% without evident fluctuations (5–100 cycles) (Fig. 4j), inferring that stable electrolyte/electrode interfaces were built up.

Design of 600 Wh kg⁻¹ Li metal pouch cells

To enable pouch cells with high energy densities (≥ 600 Wh kg⁻¹) and long cycle life (≥ 100 cycles), several critical challenges must be addressed, including the corrosion of Li metal, gassing, and poor stability of the high-Ni positive electrode, and rapid electrolyte depletion. The combination of solid-state GMFN electrolyte and BGPL@Li has proven effective in solving the above problems. To our surprise, our findings reveal that electrolyte depletion may play a more pivotal and detrimental role than the thickness of Li foil in the deterioration of our designed 500 Wh kg⁻¹-level Li metal pouch cells (and beyond) over 100 cycles. As shown in Supplementary Fig. 64, a 7 Ah-class pouch cell (parameters in Supplementary Table 6) with an electrolyte-to-capacity (E/C) ratio of 1.5 g Ah⁻¹ and using 20 μ m bare Li@Cu foil and the MFN electrolyte, was cycled at a 0.2 C/0.5 C charge/discharge rate. A sudden capacity drop was observed after 60 cycles. Upon dismantling the pouch cell, a considerable amount of unreacted fresh Li was still found beneath the accumulated byproduct of side reactions. The pouch cell was completely dry, with no residual electrolyte remaining in the aluminum plastic bag (Supplementary Fig. 65). Further increasing the injection dose of electrolyte to 2.4 g Ah⁻¹, the cycle life of the 7 Ah-class pouch cell is extended to over 100 cycles. This evidence indicates that electrolyte drainage, rather than the limited amount of Li, could be the decisive factor leading to pouch cell failure within 100 cycles (Supplementary Note 6 Part I).

Based on the above considerations, the solid-state GMFN electrolyte and the BGPL@Li were integrated into the pouch cell assembly to design an 11 Ah-class pouch cell (11.3 Ah), achieving an high specific energy of 604.2 Wh kg⁻¹ (the first-cycle voltage profile and 11 Ah-class pouch cell shown in Fig. 5a). Specific parameters of the pouch cell are shown in Fig. 5c and Supplementary Table 7 (pictures of the thickness measurement and pouch cells during cycling are shown in Supplementary Fig. 66).

Operating a 600 Wh kg⁻¹-level cell can be rather challenging, particularly under lean electrolyte conditions due to the easy drainage

of conventional liquid electrolyte triggered by reactive Li-metal and high-voltage positive electrodes (Supplementary Note 6 Part II, Supplementary Fig. 67). The as-fabricated 11 Ah-class GMFN-BGPL@Li pouch cell exhibits a high specific energy of 604.2 Wh kg⁻¹, retains 92.83% of its original specific energy after 100 cycles at 0.1 C/0.2 C charge/discharge mode (Fig. 5b), and maintains 560.9 Wh kg⁻¹ after 100 cycles (with comparison to other works summarized in Supplementary Table 7). The excellent long-term cycling stability is primarily attributed to the decelerated electrolyte depletion as well as prevented side reactions between the GMFN, the BGPL@Li, and the Ni92 positive electrode. Supplementary Fig. 68 and Supplementary Table 8 compare the pouch cell energy densities which can be obtained from recently reported works (only the “package-included” energy densities are taken into consideration)^{5,44–52}. Furthermore, another 11 Ah-class cell (with the same parametric design shown in Supplementary Table 9) is tested at a lower rate of 0.2 C/0.5 C (Supplementary Fig. 69). The increased rate places greater challenges on reaction kinetics and interfacial stability compared to that under lower rates, especially in systems utilizing thick positive electrodes (>5 mAh cm⁻²) and multi-layer LMBs. Despite these challenges, this 11 Ah-class pouch cell still demonstrates stable cycling performance, retaining 85.71% of its initial capacity after 100 cycles at 0.2 C/0.5 C.

The X-ray computed tomography (CT) is utilized to visualize the macroscopic homogeneity of pouch cells before and after cycling. Two 3 Ah-class pouch cells, each consisting of 5 positive electrodes and 6 negative electrodes, were prepared using the LP334 electrolyte with bare Li and the GMFN electrolyte with BGPL@Li, respectively, and were cycled for 30 cycles at a 0.2 C/0.5 C charge/discharge rate (Supplementary Fig. 70). Figure 5d presents the top-view CT images of these two pouch cells. The cell with the GMFN electrolyte and BGPL@Li exhibits a flat morphology, while the cell with the LP334 electrolyte and bare Li shows noticeable fluctuations, indicating an uneven stress distribution within the cell⁵³. The cross-sectional CT image of the pouch cell with the LP334 electrolyte and bare Li shows a larger thickness of 2.74 mm, compared to the 2.14 mm thickness observed in the cell with the solid-state GMFN electrolyte and BGPL@Li (Fig. 5e and Supplementary Fig. 71). The BGPL@Li-regulated Li plating, along with the reduced side reactions from the solid-state GMFN electrolyte, synergistically help minimize electrode expansion, resulting in a flatter cell morphology.

Notably, safety-related characterization is rarely reported for LMBs with energy densities of 500 Wh kg⁻¹ or higher. Two 5 Ah-class Li||Ni92 pouch cells, using the LP334 with bare Li and GMFN with BGPL@Li (7 positive electrodes + 8 negative electrodes), were cycled at 0.2 C/0.5 C for 10 cycles and then discharged to 3.65 V (state of charge: 25%). The Li||Ni92 pouch cell with the bare Li and LP334 electrolyte exploded after the penetration (Fig. 5f), while the Li||Ni92 pouch cell with the BGPL@Li and GMFN electrolyte remained intact without smoke or open flames (Supplementary Fig. 72, Supplementary Video 3). Figure 5g presents photos of the pouch cells before and after penetration. The Li||Ni92 pouch cell with the BGPL@Li and GMFN electrolyte shows no thermal runaway, and the surface temperature remains almost unchanged during the nail puncture process (Fig. 5h). The Li||Ni92 cell with the BGPL@Li and GMFN was further subjected to a nail puncture test when fully charged to 4.25 V, and it successfully passed without thermal runaway or ignition (Supplementary Fig. 73, Supplementary Video 4). Additionally, accelerating rate calorimetry (ARC) tests were performed to evaluate the thermal stability and thermal runaway behaviors of gel-solid-state GMFN-BGPL@Li pouch cell and liquid LP334-Bare Li pouch cell (Supplementary Note 7, Supplementary Figs. 74–76). The T_1 values (self-heating initiation) of LP334-Bare Li (107.2 °C) and GMFN-BGPL@Li (106.5 °C) cells are similar. However, the LP334-Bare Li pouch cell undergoes violent thermal runaway at 180.4 °C (dT/dt > 1150 °C s⁻¹), leading to the cell explosion. In contrast, the GMFN-BGPL@Li cell exhibits delayed thermal runaway

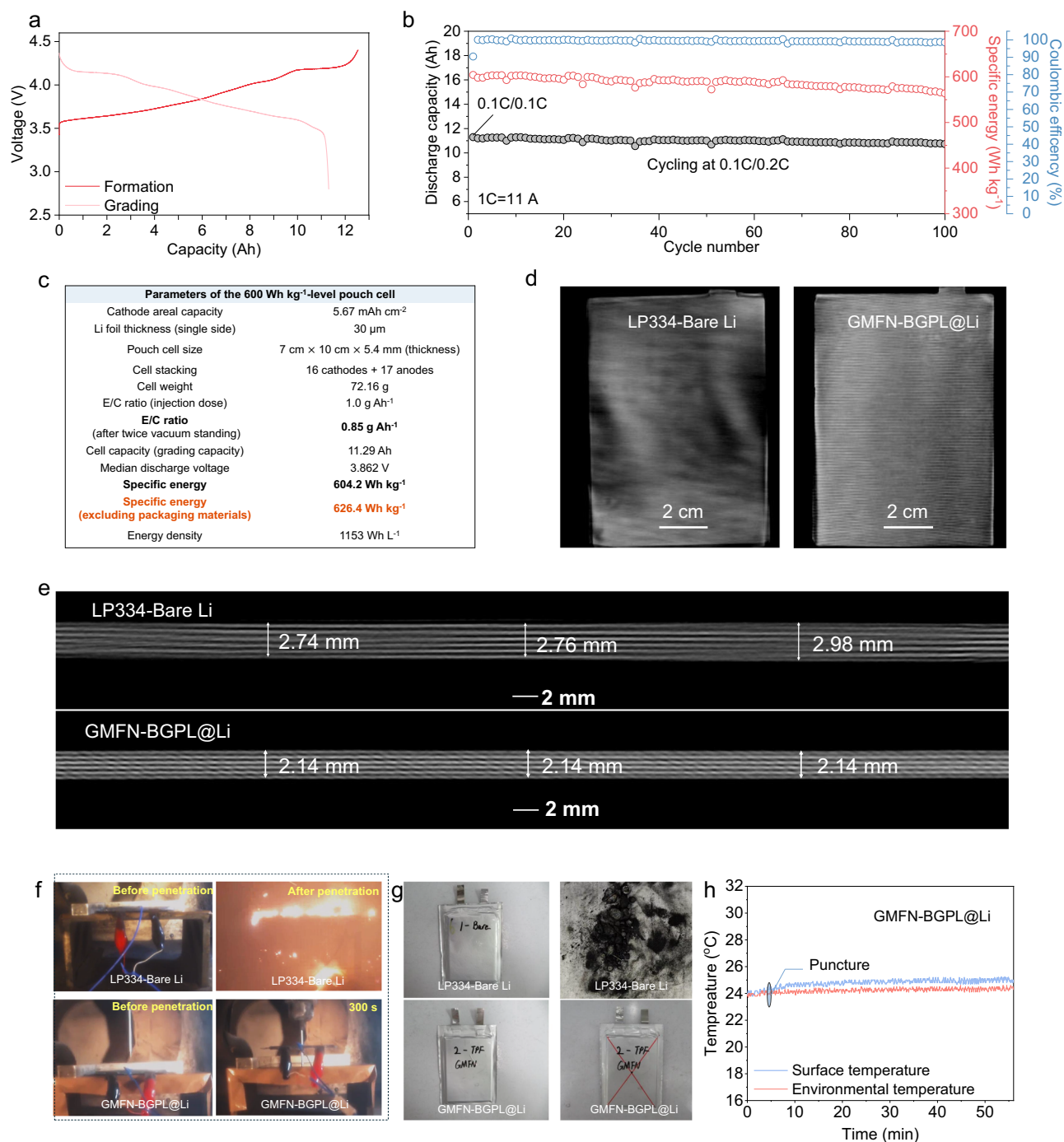


Fig. 5 | Electrochemical performance and characterization of the GMFN-enabled and BGPL@Li-assembled pouch cells. **a** Formation cycle and grading performance of the 11 Ah-class pouch cell using BGPL@Li and GMFN electrolyte. **b** Cycling performance (from 2.8 to 4.4 V) along with the corresponding CE and specific energy of the 11 Ah-class pouch cell using BGPL@Li and GMFN electrolyte. **c** Parameters of the 11 Ah-class pouch cell based on the BGPL@Li and GMFN electrolyte. The energy density is calculated based on the original pouch cell size as listed in this form. **d, e** X-ray tomography images of the cycled 3 Ah-class pouch

cells using the LP334 with the bare Li and the GMFN electrolyte with the BGPL@Li (tested at the SOC of 0%, after 30 full charge/discharge cycles at 0.2 C/0.5 C rate under 25 °C) in side view (**d**) and top view (**e**). **f, g** Pictures of the 5 Ah-class pouch cell (SOC ≈ 25%) the LP334 with the Bare Li and the GMFN electrolyte with the BGPL@Li during (**f**) and after (**g**) nail puncture tests. **h**, Temperature curve of the 5 Ah-class pouch cell using the GMFN electrolyte coupled with the BGPL@Li during the nail puncture test.

and stabilized self-heating (0.6 °C min⁻¹) between 141 and 190 °C due to the endothermic gel-solid-state electrolyte. This demonstrates that the cell using the GMFN and BGPL@Li shows a higher safety performance and is more likely to avoid thermal runaways under extreme conditions, compared to the LMBs with conventional liquid electrolyte and unmodified Li foils.

In summary, emerging maritime and aerial electric transport applications urge batteries with higher specific energies (≥400 Wh kg⁻¹). However, the current specific energy of Li-ion batteries falls short of the requirements. In this work, we combine electrolyte engineering and surface modification of Li metal, as well as advanced extreme battery design, to enable the scalable fabrication of

pouch-type energy-dense ($\geq 600 \text{ Wh kg}^{-1}$) and long-cycling (≥ 100 cycles) LMBs. The in-situ formed multifunctional polymer scaffold networks effectively confine unstable free carbonate solvents, suppress electrolyte-induced parasitic reactions, mitigate gassing issues, and ultimately enhance electrode stability and reduce electrolyte decomposition. Meanwhile, a thin, uniform, and dense BGPL layer is designed to improve Li stripping/plating reversibility (CE: 99.66%) in carbonate electrolytes while enhancing resistance to moisture and oxygen. The effective surface modification of BGPL enables damage-free processing of Li metal, eliminating the harsh environmental and equipment requirements typically associated with large-scale production of LMBs. The BGPL@Li remains stable under a dew point of -30°C for over 10 days, with more than 96% of the active Li preserved, while the bare Li foil completely decomposes under the same conditions. The synergistic implementation of the solid-state GMFN electrolyte and BGPL@Li leads to the development of 11 Ah-class LMBs with a specific energy of 604.2 Wh kg^{-1} (626.4 Wh kg^{-1} excluding packaging materials) and long cycle life, showing an energy retention of 92.83% after 100 cycles. This marks a milestone in the development of rechargeable LMBs and provides valuable insights into the future design of high-specific-energy batteries.

Methods

Molecule dynamic simulations

Molecule dynamics (MD) simulations were conducted to explore the microscopic interactions and dynamic properties of electrolyte systems in the presence and absence of the polymer skeleton PETEA-HFBMA-DAP (1:1:1 tri-block structure, with block structures illustrated in Supplementary Fig. 2). Two distinct electrolyte models were built for comparison. The first electrolyte model incorporated Li^+ (10), FEC (60), PF_6^- (10), DMC (80), EMC (30), FEMC (30), and PETEA-HFBMA-DAP (5) to mimic the gel-solid-state electrolyte environment (Supplementary Data 1). While the second one consisted of Li^+ (10), FEC (60), PF_6^- (10), DMC (10), EMC (80), and FEMC (30) to represent the liquid electrolyte environment (Supplementary Data 2).

Amorphous initial configurations were constructed via the Amorphous Cell module in Materials Studio, utilizing a Monte Carlo packing algorithm to randomly place all molecules in a 3D periodic simulation box. Periodic boundary conditions (PBCs) were imposed along all three directions to mimic bulk electrolyte behavior. To release unfavorable local stresses and attain stable structures, energy minimization was first carried out, followed by MD simulations in the Forcite module⁵⁴. The systems were equilibrated via a 200 ps NPT simulation at 298 K and 0.1 MPa to adjust the density and relax the volume. Subsequently, a production run was performed in the NVT ensemble for 20 ns, regulated by a Nose–Hoover thermostat with a Q ratio of 0.01.

The intermolecular interactions were modeled using the COMPASS III force field, with charges assigned based on the force field parameters. Specifically, Van der Waals interactions were evaluated using an atom-based summation method, while long-range electrostatic effects were handled by Ewald summation. A time step of 1 fs was adopted for numerical integration, and trajectory frames were recorded at intervals of 5000 steps for further analysis. The binding energy (E_{bind}) between two components was evaluated to quantitatively assess their interaction strength, using the following established expression:

$$E_{\text{bind}} = E_{A+B} - (E_A + E_B) \quad (1)$$

where E_{A+B} is the total potential energy of the complex formed by components A and B (with other species excluded), and E_A and E_B are the energies of the isolated components A and B, respectively³⁵. As an illustration, for the interaction between FEC and the polymer skeleton (PETEA-HFBMA-DAP), $E_{\text{FEC} + \text{polymer skeleton}}$ corresponds to the total

energy of the FEC and the polymer skeleton, while E_{FEC} and $E_{\text{polymer skeleton}}$ represent the energies of isolated FEC and polymer skeleton, respectively. A more negative E_{bind} value indicates a stronger binding interaction between the two species.

The coordination number of species i around a central species j within the first solvation shell is calculated as follows:

$$N_i = 4\pi n_j \int_0^{R_M} g_{ij}(r) r^2 dr \quad (2)$$

In which, R_M is the distance of the first minimum following the first peak in the radial distribution function (RDF), $g_{ij}(r)$, which is a standard approach for bulk liquid. All molecular structures and trajectories were visualized using VMD software.

Materials

LP334 electrolyte (1 M LiPF_6 in EC/DEC/DMC (v/v/v = 3:3:4) + 2 vol% FEC + 2 vol% vinylene carbonate), lithium hexafluorophosphate (LiPF_6 , >99.9% purity), ethyl methyl carbonate (EMC, >99.9% purity), dimethyl carbonate (DMC, >99.9% purity), methyl (2,2,2-trifluoroethyl) carbonate (FEMC, >99% purity), fluoroethylene carbonate (FEC, >99.9% purity), LiDFOB salt (>99.5% purity) and lithium nitrate (LiNO_3 , >99.9% purity) salt are purchased from DodoChem (Suzhou) Co., Ltd. PETEA (>99% purity), DAP (97% purity) and HFBMA (98% purity) are purchased from Sigma-Aldrich. Magnesium bis(trifluoromethanesulfonimide) ($\text{Mg}(\text{TFSI})_2$, 97% purity), HMDS ($\geq 99\%$ purity), benzoyl peroxide (BPO, >99.9%) and poly(vinylidene fluoride-co-hexafluoropropylene) (PVDF-co-HFP) are purchased from the Aladdin (Shanghai) Co., Ltd. The 1,3,5-Trioxane (TX) is purchased from the TCI (Shanghai) Co., Ltd. The single crystalline Ni92 is provided by the BTR Co., Ltd. In this work, the unit “M” stands for molar salt in a liter solvent. The LATP and cellulose coated separator (9 μm thick, 110 mm wide, 500 m long, with a porosity of 37%) used for pouch cell assembly was custom-made by Shenzhen Senior Technology Material Co., Ltd.

Gel-solid-state electrolyte preparation

The MFN and MN electrolytes are prepared by mixing all the contained components under 60°C in an Ar-filled glove box. The MFN electrolyte (M: denoted as $\text{Mg}(\text{TFSI})_2$; F: denoted as FEMC; N: denoted as LiNO_3) is prepared by adding 12 mmol LiPF_6 and 0.2 mmol $\text{Mg}(\text{TFSI})_2$ are dissolved with a 10 mL mixture of FEC/FEMC/EMC/DMC (v/v/v/v = 3: 1.5: 1.5: 4) under stirring at 60°C , followed by the addition of 0.65 mmol LiNO_3 into the solution. After the electrolyte solution turns transparent, 0.1 wt% HMDS is added, and the MFN electrolyte is acquired. The MN electrolyte (1.2 M LiPF_6 in FEC/EMC/DMC (v/v/v = 3: 3: 4) + 20 mM $\text{Mg}(\text{TFSI})_2$ + 65 mM LiNO_3 + 0.1 wt% HMDS) is prepared in the same way. The GMFN electrolyte for the coin cells is prepared by adding 1.5 wt% PETEA, 1.5 wt% DAP, 1 wt% HFBMA, and 0.2 wt% BPO into the MFN electrolyte in sequence, followed by 4 h heating in the vacuum oven under 65°C . For the pouch cells, the GMFN electrolyte is heated in the oven under 60°C for only 1 h.

Characterizations

Before the post-mortem characterizations included in this work, all cycled negative electrodes and positive electrodes were washed with 1,2-dimethoxyethane and dried for over 6 h under vacuum. A scanning electron microscope (SEM), TESCAN MIRA 4, was used to acquire the morphologies of the materials. Time-of-flight Secondary Ion Mass Spectrometer (TOF-SIMS) tests with high ion detection sensitivity were carried out by TOF-SIMS V (ION-TOF GmbH) and analyzed by ION-TOF software. The X-ray photoemission spectroscopy (XPS) test was performed by the Thermo K-alpha X-ray Photoelectron Spectrometer System. The NMR spectra were acquired through the Bruker Avance NEO 600 and Varian VNMRs 600 MHz. The WAXS characterization is performed using Xenocs Xeuss 3.0. The scanning transmission

electron microscopy (STEM) images were acquired using the FEI Tecnai F20 microscope. Ionic conductivities of electrolytes were measured based on the CR2032 symmetric coin cell configuration using stainless steel, while the Li-ion transference number was tested utilizing the Li foils. The X-ray computed tomography (CT) test was performed using the ZEISS Xradia 520 on the cycled 3 Ah-class pouch cells after 30 cycles. The nail puncture test was conducted with a steel (diameter of 3 mm) and a tilt angle of 60° at the rate of 10 mm s⁻¹.

Cell assembly and electrochemical performance

All the coin cells are assembled in the Argon-filled glove box with water and oxygen content lower than 0.1 ppm. The Li||Cu asymmetric half cells are assembled (with 30 µL of electrolyte injected) using the Li foil (250 µm, diameter: 12 mm) and Cu foil (diameter: 16 mm) for determining plating/stripping efficiencies of LP334 and MFN electrolytes by the Aurbach method, where the reservoir capacity of 5 mAh cm⁻², cycling current 0.5 mA cm⁻², and plating/stripping capacity of 1 mAh cm⁻² are adopted. To determine the plating/stripping efficiency of the BGPL@Li, Li||BGPL@Li@Cu asymmetric cells are assembled using the Li foil (250 µm, diameter: 16 mm) and BGPL@Li@Cu foil (50 µm, diameter: 12 mm). Initially, the Li on BGPL@Li@Cu is stripped by charging the cell to 1.0 V at 0.5 mA cm⁻², and then the same parameters are adopted for the Aurbach method-based measurement as the Li||Cu cell test. The Al-coated stainless-steel cases with an additional Al metal foil (0.5 mm thickness) are placed at the positive electrode side for all coin cells with Ni92 and NCM811 positive electrodes in this work. The positive electrode slurry for coin-cell tests is prepared using the THINKY ARE-250 CE mixer at 2000 rpm for 30 min. All the positive electrodes are prepared through blade-cast method (positive electrode active materials: super P: CNT: PVDF = 97: 1.45: 0.05: 1.5), followed by twice calendar roll processes, and then punched into disks (ϕ = 12 mm), with areal capacity around 2.9 mAh cm⁻² for cell performance evaluation. The single crystalline Ni92 is provided by the BTR Co., Ltd. The CNT slurry is purchased from Jiangsu Cnano Technology Co., Ltd. The PVDF is purchased from the Solway (the 5130 type). The single-side coated high-loading Ni92 positive electrode for the coin cell test (5.5 mAh cm⁻²) is prepared using the pilot scale equipment (Fig. 4i). The double-side coated Ni92 positive electrode with a preset loading of 26 mg cm⁻² is used for all pouch cells. Pouch cells with different numbers of staking layers are prepared using automatic stacking equipment. All Li||Ni92 coin cells were assembled with an electrolyte volume of 60 µL.

3 Ah-class (5 positive electrodes and 6 negative electrodes) and 5 Ah-class pouch cells (7 positive electrodes and 8 negative electrodes) are used for the CT characterization and the nail puncture test respectively, based on bare Li or BGPL@Li foils (100 µm in total; 50 µm for single side). The module powering the UAV consists of 4-series 3 Ah-class (5 positive electrodes and 6 negative electrodes) pouch cells with the GMFN electrolyte. 7 Ah-class pouch cells are prepared with unmodified Li@Cu composite foils (20 µm for the single side, 20 × 2 µm in total) for determining the decisive factor (electrolyte amount *vs.* lithium thickness) in our designed full-cell pouch batteries. The 11 Ah-class (600 Wh kg⁻¹-level) pouch cell is prepared with 16 positive electrodes and 17 BGPL@Li foils (60 µm thick before polymer coating). The energy density (Wh L⁻¹) is calculated based on the calculated volume of pouch cell size (7 cm × 10 cm × 5.4 mm). The preserved amount of electrolyte is determined by measuring the weight loss of the pouch cells after two sufficient vacuum standing procedures. All the pouch cells were tested under a pressure of 360 kPa using the pressure sensor module (purchased from Guangdong Canrd New Energy Technology Co., Ltd.), with another film pressure sensor employed for pressure calibration.

The floating analysis is based on the Li||Ni92 cell, which is first charged to 4.3 V and then applied with the voltage staircase from 4.4 V to 5.2 V (interval of 0.1 V) for 5 h at each voltage or the current reaches

the lower limit of 0.005 mA cm⁻². The electrochemical impedance spectroscopy (EIS) measurements were conducted in potentiostatic mode, with a frequency range of 100 kHz to 0.1 Hz, an amplitude of 10 mV, and 30 data points recorded per decade. Measurements were performed under open-circuit voltage after the cells were allowed to stand for at least 6 h before the EIS measurement. Tafel plots were measured from -0.2 V to 0.2 V at a scan rate of 0.2 mV s⁻¹. The LSV curves, EIS, and Tafel plots were all tested using the Princeton PAR-STAT (PMC 1000) electrochemical workstation.

Manufacturing of the BGPL@Li and other modified Li foils

The BGPL@Li solution is prepared by dissolving 3.6 g 1,3,5-Trioxane/Trioxin (TX), 1.8 g PVDF-co-HFP and 0.036 g LiDFOB in 10 mL dimethyl carbonate (DMC) under 70 °C. The PVDF-co-HFP modified Li foil and TX-modified Li foil modified are prepared by dissolving 1.8 g PVDF-co-HFP, 0.036 g LiDFOB and 3.6 g TX, 0.036 g LiDFOB in 10 mL DMC, respectively. The pilot-scale manufacturing of PVDF-co-HFP is prepared with the MTI MSK-AFA-EI300 film coating machine. A polyethylene terephthalate (PET) film is loaded upon the bare Li before the coating process is started. For the Li@Cu foils, they are loaded upon the 6 µm-Cu foil (width of 280 mm) with tapes as the supporter during the blade casting process. For the pure Li foils, they are loaded upon two layers of 6 µm-Cu foils (width of 200 mm and 280 mm, respectively) for the prevention of foil breakage during roll-to-roll tape-casting. The temperature of the oven during the coating process is 80 °C with a feed rate of 100 mm min⁻¹.

Data availability

The authors declare that all the data supporting the findings of this study are available within the main text and its supplementary information files. Source data are provided with this paper.

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T. Zhao and Y. Li supervised the work. X. Peng and Y. Li conducted the concept design. X. Peng, Y. Zhang, L. Chen, N. Chen, Y. Jiang, Z. Yang, and H. Pan provided material characterizations. X. Peng, Y. Zhang, and H. Li prepared the electrode materials and conducted the experiments. X. Peng and Q. Wang conducted the MD simulations. X. Peng and Y. Li analyzed the results and drafted the manuscript. X. Peng, Y. Li, and T. Zhao wrote and revised the manuscript. All authors commented on the final manuscript. X. Peng and Y. Zhang contributed equally to this work.

Competing interests

X. Peng was an employee, and Y. Zhang, H. Li, and Q. Wang are employees of the Research Institute of Renewable Energy and Advanced Materials, Zijin Mining Group Co., Ltd., Changsha, Hunan 410013, China. The other authors declare no competing interests.

Additional information

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